INTERNSHIP REPORT

On

CYBERSECURITY FLUENCY PATHWAY

By

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CERTIFICATE

This is to certify that the internship report entitled **"CYBERSECURITY FLUENCY PATHWAY – IBM SKILLSBUILD"** is a Bonafide record of work carried out by **SUMANTH REDDY (2023002574)** submitted in partial fulfilment of requirement for the award of degree of Bachelor of Technology in Computer Science and Engineering.

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ABOUT THE INTERNSHIP COURSE

This internship was carried out through IBM SkillsBuild, IBM's free, tuition-free digital learning platform that offers self-paced and guided courses in emerging technology domains such as Artificial Intelligence, Data Analytics, Cloud Computing, and Cybersecurity. The specific learning plan undertaken was the “Cybersecurity Fluency Pathway,” a structured pathway comprising six stackable microcredentials designed to build a solid, practical foundation in cybersecurity concepts, tools, and career pathways.

The pathway is intended for both newcomers to the field and professionals seeking to strengthen their knowledge. It combines conceptual instruction with hands-on activities involving security tools, frameworks, and real-world inspired scenarios, allowing learners to apply what they study rather than simply reading about it.

On successful completion of all modules and assessments, IBM issues a verifiable digital credential through Credly, confirming the learner's demonstrated understanding of cybersecurity threats, defensive principles, and operational practices.

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ABSTRACT

Cybersecurity has become one of the most critical concerns for individuals, businesses, and governments as digital infrastructure continues to expand across every sector. This report documents a one-month internship undertaken through IBM SkillsBuild's Cybersecurity Fluency Pathway, a self-paced online learning program consisting of six microcredential modules covering the foundational and applied aspects of cybersecurity.

The internship covered core topics including the CIA triad and security principles, common cyber threats and attack vectors, social engineering techniques, identity and access management, endpoint security, cryptographic methods for protecting data, cloud infrastructure security, and the operational practices used by Security Operations Centers (SOCs) to detect, respond to, and recover from incidents. In addition to conceptual study, the pathway included interactive labs, scenario-based exercises, and self-assessments that reinforced how these concepts are applied in real organizational settings.

The report presents a chapter-wise account of each topic studied, the tools and platforms used during the learning process, the outcomes and skills gained, and a reflection on how this knowledge applies to real-world cybersecurity roles. The internship concluded with the successful completion of all required modules and the award of the IBM digital credential for the Cybersecurity Fluency Pathway.

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1. INTRODUCTION

1.1 Background

Every industry today depends on digital systems to store data, run operations, and communicate with customers. This dependence has made the protection of digital assets a critical priority, and organizations across the world are actively seeking professionals with cybersecurity skills. To build a foundation in this domain, I undertook a one-month internship centered on IBM SkillsBuild's Cybersecurity Fluency Pathway, a free, self-paced learning program offered by IBM.

1.2 About IBM SkillsBuild

IBM SkillsBuild is a free, global education program that provides individuals with job-relevant skills in high-demand fields such as Artificial Intelligence, Cloud Computing, Data Analytics, and Cybersecurity. Courses are designed by IBM subject-matter experts and combine short video lessons, readings, interactive labs, and assessments. Learners who complete a pathway earn an IBM digital credential, verified and issued through Credly, which can be shared on professional platforms such as LinkedIn.

1.3 Objectives of the Internship

- To build a strong conceptual foundation in cybersecurity principles and terminology.
- To understand common categories of cyber threats and how attackers exploit vulnerabilities.
- To learn the practices organizations use to protect identities, endpoints, networks, and cloud infrastructure.
- To understand how security operations teams detect, respond to, and recover from incidents.
- To explore career pathways and roles available in the cybersecurity industry.

1.4 Scope

The internship was limited to the modules covered under the Cybersecurity Fluency Pathway on IBM SkillsBuild. It did not involve deployment on live organizational infrastructure; instead, it relied on IBM's simulated labs, case studies, and guided exercises to build applied understanding within a controlled learning environment.

Table 1.1: Internship Timeline

Week	Focus Area	Key Outcome
Week 1	Cybersecurity fundamentals, CIA triad, security domains	Conceptual foundation established
Week 2	Cyber threats, attack types, social engineering	Threat awareness developed
Week 3	IAM, endpoint security, cryptography	Access control & data protection understood
Week 4	Cloud security, SOC operations, incident response, capstone assessment	Pathway completed, credential earned

2. CYBERSECURITY FUNDAMENTALS

2.1 What is Cybersecurity

Cybersecurity is the practice of protecting systems, networks, applications, and data from unauthorized access, disruption, or damage. It encompasses the technologies, processes, and controls designed to defend digital assets against an ever-evolving landscape of threats, ranging from individual hackers to organized cybercrime groups and nation-state actors.

2.2 The CIA Triad

The CIA triad — Confidentiality, Integrity, and Availability — forms the foundational model that guides most cybersecurity decisions and controls.

- Confidentiality: Ensuring that information is accessible only to authorized individuals, typically enforced through encryption, access controls, and authentication.
- Integrity: Ensuring that data remains accurate and unaltered except by authorized action, often protected using hashing and checksums.
- Availability: Ensuring that systems and data are accessible to authorized users when needed, supported by redundancy, backups, and disaster recovery planning.

(Figure 2.1: The CIA Triad — illustrating the three pillars of confidentiality, integrity, and availability that anchor all cybersecurity strategy.)

2.3 Security Domains

The pathway introduced the major domains that make up an organization's overall security posture, including network security, application security, information security, operational security, disaster recovery and business continuity planning, and end-user education. Each domain addresses a different layer of risk, and effective cybersecurity strategy requires coordinated controls across all of them rather than relying on a single line of defense.

3. CYBER THREATS AND ATTACK TYPES

3.1 Threat Actors and Threat Groups

The pathway categorized threat actors by motivation and capability: script kiddies with limited skill exploiting known vulnerabilities, hacktivists motivated by ideology, organized cybercriminal groups driven by financial gain, and advanced persistent threat (APT) groups, often state-sponsored, capable of long-term, stealthy campaigns against high-value targets.

3.2 Common Attack Types

- Malware: Malicious software such as viruses, worms, trojans, and ransomware designed to damage systems or hold data hostage.
- Phishing: Fraudulent communications, usually email, designed to trick users into revealing credentials or installing malware.
- Denial-of-Service (DoS/DDoS): Attacks that overwhelm systems with traffic to make services unavailable to legitimate users.
- Man-in-the-Middle (MitM): Interception of communication between two parties to steal or alter data in transit.
- SQL Injection and Cross-Site Scripting (XSS): Application-layer attacks that exploit poorly validated input to manipulate databases or inject malicious scripts.

(Figure 3.1: Common Cyberattack Categories — a summary map of the major attack vectors covered in the pathway.)

3.3 Social Engineering

A significant portion of the pathway focused on social engineering — the manipulation of human psychology rather than technical vulnerabilities to gain unauthorized access. Techniques studied included pretexting, baiting, tailgating, and business email compromise. Case studies illustrated how even well-secured technical systems can be compromised when employees are not adequately trained to recognize manipulation tactics, reinforcing that people are often the weakest link in the security chain.

4. IDENTITY, ACCESS MANAGEMENT AND ENDPOINT SECURITY

4.1 Identity and Access Management (IAM)

IAM refers to the frameworks and technologies used to ensure that the right individuals have access to the right resources at the right time, for the right reasons. Core concepts studied included authentication (verifying identity), authorization (granting permissions), the principle of least privilege, multi-factor authentication (MFA), and single sign-on (SSO).

(Figure 4.1: Identity and Access Management Cycle — showing the flow from identification through authentication, authorization, and continuous monitoring.)

4.2 Endpoint Security

Endpoints — laptops, mobile devices, servers, and IoT devices — represent common entry points for attackers. The pathway covered endpoint protection strategies including antivirus and endpoint detection and response (EDR) tools, patch management, device encryption, and mobile device management (MDM) policies that allow organizations to secure a growing and increasingly remote workforce.

5. CRYPTOGRAPHY AND DATA PROTECTION

5.1 Why Cryptography Matters

Cryptography underpins the confidentiality and integrity pillars of the CIA triad. It converts readable data (plaintext) into an unreadable form (ciphertext) that can only be reversed by authorized parties holding the correct key, protecting data both at rest and in transit.

5.2 Types of Cryptography

The pathway distinguished between symmetric encryption, where the same key is used to encrypt and decrypt data (e.g., AES), and asymmetric encryption, which uses a public-private key pair (e.g., RSA). Hashing algorithms such as SHA-256, which produce a fixed-length digest used to verify data integrity, were also covered, along with the role of digital certificates and Public Key Infrastructure (PKI) in enabling secure communication over the internet (HTTPS/TLS).

Table 5.1: Common Cryptographic Algorithms

Algorithm	Type	Typical Use
AES	Symmetric	Encrypting stored data / files
RSA	Asymmetric	Secure key exchange, digital signatures
SHA-256	Hashing	Data integrity verification
TLS/SSL	Protocol	Securing data in transit (HTTPS)

6. CLOUD SECURITY AND SECURITY OPERATIONS

6.1 Cloud Infrastructure Security

As organizations migrate workloads to public, private, and hybrid cloud environments, securing that infrastructure has become essential. The pathway covered the shared responsibility model, which defines which security tasks are handled by the cloud provider versus the customer, along with cloud-specific risks such as misconfigured storage, insecure APIs, and inadequate identity controls.

6.2 Security Operations Centers (SOC)

A Security Operations Center is the team and facility responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity incidents. The pathway explained the tools SOC analysts use, including Security Information and Event Management (SIEM) platforms, and the tiered structure of SOC roles from Tier 1 alert triage to Tier 3 threat hunting.

6.3 Incident Response Lifecycle

A structured incident response process helps organizations minimize damage and recover quickly from security events. The pathway used a standard lifecycle model:

- Preparation: Establishing policies, tools, and training before an incident occurs.
- Identification: Detecting and confirming that a security incident has occurred.
- Containment: Limiting the spread and impact of the incident.
- Eradication: Removing the root cause, such as malware or unauthorized access.
- Recovery: Restoring affected systems to normal operation.
- Lessons Learned: Reviewing the incident to strengthen future defenses.

(Figure 6.1: Incident Response Lifecycle — illustrating the six-stage cycle organizations follow when responding to a security incident.)

7. TOOLS AND PLATFORMS EXPLORED

The internship was delivered primarily through the IBM SkillsBuild platform (skills.yourlearning.ibm.com), which hosts self-paced video lessons, readings, knowledge checks, and interactive lab simulations. The table below summarizes the categories of tools and resources engaged with during the pathway.

Table 7.1: Cybersecurity Tools Explored

Category	Examples	Purpose
Learning Platform	IBM SkillsBuild portal	Course delivery, labs, assessments
Credentialing	Credly	Issuing and verifying digital badges
Threat Reference	MITRE ATT&CK (referenced)	Classifying attacker tactics & techniques
Monitoring Concepts	SIEM platforms (conceptual)	Centralized log analysis & alerting

In addition to platform-native simulations, the pathway referenced industry-standard tools and frameworks that professionals use in practice, such as SIEM platforms for log correlation, vulnerability scanners for identifying weaknesses in systems, and frameworks like the MITRE ATT&CK matrix for classifying adversary behavior.

8. RESULTS AND LEARNING OUTCOMES

By the end of the internship, all modules of the Cybersecurity Fluency Pathway were completed and the associated assessments were passed, resulting in the successful award of the IBM SkillsBuild digital credential (Plan ID: PLAN-EC6B83791960) on 20 June 2026.

Key skills and outcomes gained during the internship include:

- A working understanding of the CIA triad and how it guides cybersecurity decision-making.
- The ability to identify and describe common cyberattack types and social engineering tactics.
- Familiarity with identity and access management principles, including MFA and least privilege.
- A conceptual grasp of symmetric and asymmetric cryptography and their real-world applications.
- Awareness of cloud security responsibilities and the structure and function of a SOC.
- Understanding of the six-stage incident response lifecycle used by security teams.
- Exposure to the range of career roles available in cybersecurity and the certifications associated with them.

(Figure 9.1: IBM SkillsBuild Completion Certificate — verifying successful completion of the Cybersecurity Fluency Pathway.)

9. CONCLUSION

This one-month internship through IBM SkillsBuild's Cybersecurity Fluency Pathway provided a comprehensive introduction to the field of cybersecurity, blending theoretical concepts with applied, scenario-based learning. Starting from foundational principles such as the CIA triad, the internship progressed through threat identification, identity and access management, cryptography, cloud security, and the operational processes organizations rely on to detect and respond to incidents.

The structured, self-paced nature of the pathway made it possible to build meaningful competency within a short timeframe, while the interactive labs ensured that concepts were not just memorized but understood in context. This internship has strengthened my interest in pursuing a career in cybersecurity and has provided a solid base of knowledge on which to build more advanced, specialized skills such as penetration testing, digital forensics, or security operations, going forward.

REFERENCES

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4. MITRE ATT&CK Framework, https://attack.mitre.org/.

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APPENDIX – IBM COMPLETION CERTIFICATE

7/22/26, 9:33 AM Completion Certificate | SkillsBuild

IBM **SkillsBuild**

Completion Certificate

This certificate is presented to

sumanth reddy

for the completion of

Cybersecurity Fluency Pathway (Earn 6 microcredentials!)

(PLAN-EC6B83791960)

According to the Your Learning Builder - Plans system of record

Completion date: 20 Jun 2026 (GMT)

https://skills.yourlearning.ibm.com/certificate/PLAN-EC6B83791960

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